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PROBLEM IN ONE DIMENSION CHAP. 5

5-10 Hermite polynomials. The n th Hermite polynomial $H_n(x)$ is a solution of the differential equation

$$H_n'' - 2xH_n' + 2nH_n = 0$$

it is a polynomial of degree n in which the coefficient of the term in x^n is 2^n . By differentiation of Eq. (5-116), we obtain

$$(H_n')'' - 2x(H_n')' - 2H_n' + 2nH_n' = 0$$

whence the function $H_n'(x) = dH_n/dx$ satisfies the differential equation

$$\phi'' - 2x\phi' + 2(n-1)\phi' = 0$$

This equation, however, is also satisfied by H_{n-1} , and since it has only one polynomial solution, we conclude that

$$H_n' = CH_{n-1}.$$

By comparison of the terms in x^{n-1} , C can be evaluated:

$$n2^n x^{n-1} = C2^{n-1} x^{n-1}$$

The constant C is therefore $2n$, and

$$H_n' = 2nH_{n-1},$$

which is a recurrence formula. By further differentiation and manipulation of Eqs. (5-120) and (5-116), the relations

$$\begin{aligned} H_{n+1} - 2xH_n + 2nH_{n-1} &= 0, \\ H_{n+1} &= 2xH_n - H_n' \end{aligned}$$

can be derived (Problem 5-27). Equation (5-122) can be used to construct the polynomials in consecutive order, beginning with $H_0 = 1$.

The generating function, defined by

$$g(x, h) = \sum_{n=0}^{\infty} H_n(x) \frac{h^n}{n!},$$

is of great usefulness in calculations that involve the Hermite functions. By means of the recurrence formulae for H_n , a closed expression for $g(x, h)$ can be derived. Differentiating partially with respect to h , one obtains